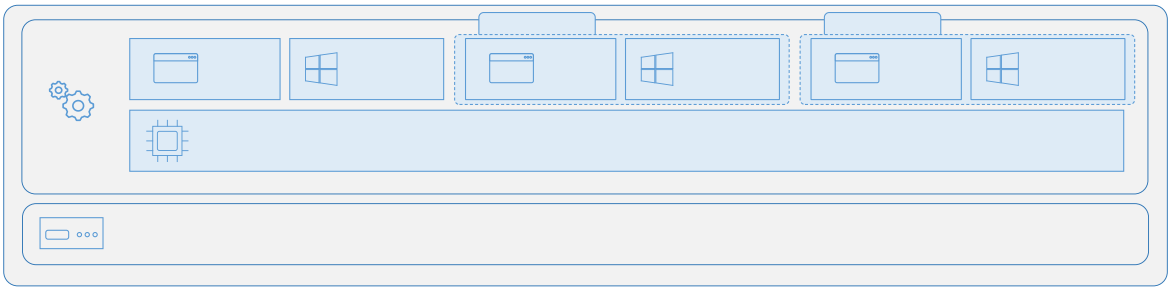
Overview of Containers in Server-2019-21

Containers are a technology for packaging and running Windows and Linux applications across different environments, both on-premises and in the cloud. They provide a lightweight, isolated environment that makes applications easier to develop, deploy, and manage.

- Containers build on top of the host operating system's kernel. They start and stop quickly, making them ideal for applications that need to rapidly adapt to changing demand.
- Their lightweight nature also allows higher density and better utilization of infrastructure resources compared to virtual machines.

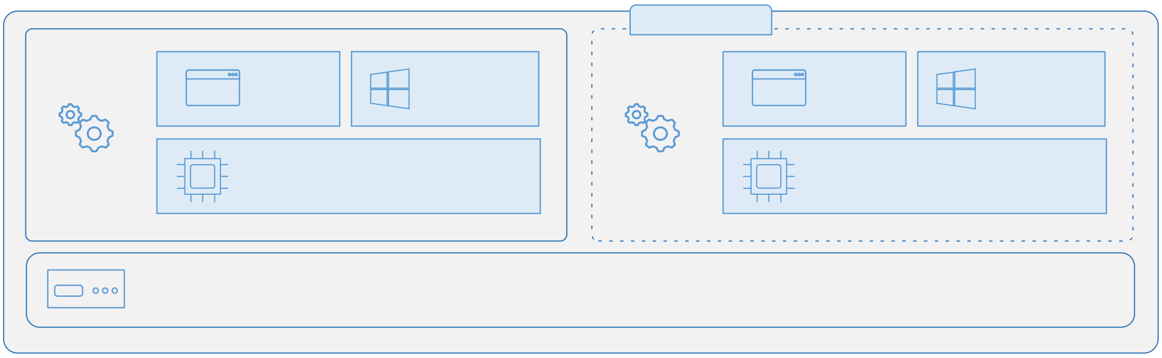
Containers enable faster deployment, easier scaling, and consistent application environments across on-premises and cloud. They use fewer resources than virtual machines, improving efficiency and portability.

How containers work

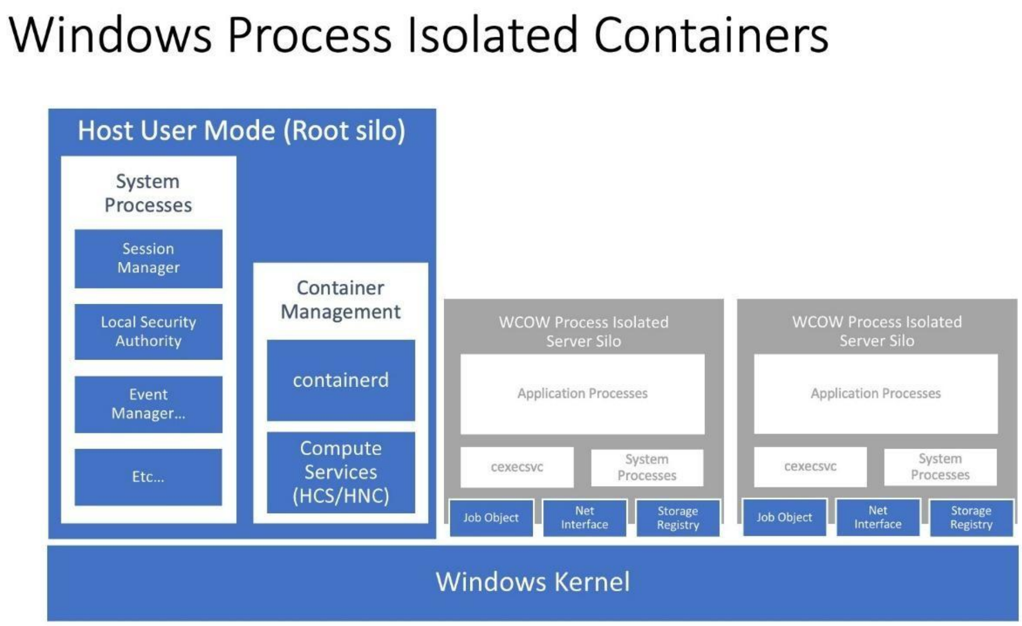


- A container is an isolated, lightweight package for running an application on the host operating system.
- Containers share the host OS kernel but do not get unrestricted access to it.
- Containers have an isolated or virtualized view of system resources.
- A container can access a virtualized version of the file system and registry, but changes affect only the container and are discarded when it stops.
- The kernel does not provide all APIs and services needed by apps. These come from system files (libraries) in user mode.
- Because a container is isolated from the host's user mode, it requires its own copy of system files.These system files are packaged into a **base image**.
- The base image provides the foundational layer and operating system services required for the container to run.

Containers vs. virtual machines



- **Operating System**: VMs run a full OS including the kernel; containers run only user-mode services with the host kernel, saving resources.
- **Compatibility**: VMs can run almost any OS; containers must run the same OS version as the host (unless using Hyper-V isolation).
- **Storage**: VMs use virtual hard disks (VHDs); containers use Azure Disks or Azure Files for storage.
- **Load Balancing**: VMs move between servers in a cluster; containers are restarted or redistributed by orchestrators.
- **Networking**: VMs use fully virtualized network adapters; containers use an isolated view of the host's network adapter with shared firewall rules.



- Containers share the same Windows kernel but run in isolated silos.
- Each silo has its own system and app processes, separate from the host.
- This gives lightweight isolation compared to full virtual machines.